Silvio Mazzucco

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Experience

- **Huawei Zurich Research Center** — Computer Vision Research Engineer
  Bayer Lab (Computer Vision) - Computational Photography — *Zürich, August 2024 - current*
  - **Product**: As part of the Computational Video team, I am in charge of the development of the preview (online) stabilization algorithm:
    - **Team Award**: Director's Award for Technical Innovation 2025.
    - **Delivered**: Successfully delivered a newly designed solution for preview stabilization for the Mate 90 Series (Mate 90/90 Pro/90 Pro Max/Ultimate) in October 2026.
    - **Individual Award**: Director's Award for Technical Innovation 2026.
  - **Research**: Conduct research on topics such as 3D/4D reconstruction and video inpainting/outpainting/controllable camera through diffusion models.
- **Computer Vision Lab @ETH** — Master Thesis - Researcher
  Source-Free Adaptation in Open-Vocabulary Semantic Segmentation — *Zürich, October 2023 - May 2024*
  - **Research**: Developed a new method for source-free adaptation of Vision-Language Models (VLMs). This was done through the introduction of a completely new vocabulary alignment strategy plus LoRA layers and a top-k selection mechanism to improve efficiency.
- **CONXAI** — Intern
  AI Product Development — *Munich, April 2023 - October 2023*
  - **Product**: Working on real AI projects (POCs and MVPs) related to the construction field varying from multi-stage classification to object detection and action recognition (delivering solutions for companies such as HILTI, PERI, Terna).
  - **Hackathon**: Led and developed the AI side of a project which won the first prize for the hackathon organized by HILTI on June 2023. The project focused on exploiting state of the art model BLIP-2 for image tagging and caption generation.
- **Autonomous System Lab @ETH** — Semester Thesis - Researcher
  3-D aware Superpixel Segmentation — *Zürich, Sept 2022 - March 2023*
  - **Research**: Research project in the field of 3D Scene Understanding; I focused on exploring the advantages and improvements coming from implementing superpixels in a semantic-NeRF model for 3-D and 2-D semantic segmentation.
- **Swissloop Tunneling** — Engineer Member
  Control System Engineer - Soil Removal Subsystem — *Zürich, Dec 2021 - Sept 2022*
  - **Sensor Interfacing**: Communication with sensors and actuators through Raspberry Pi Pico and written code in MicroPython for reading values from sensors and setting target values for actuators.
  - **Control Design and Implementation**: Controlled actuator (DC motor) through software and worked on communicating through CANOpen to employ Python written control code.

Education

- **ETH** — Zürich, Switzerland
  Master of Science in Robotics, Systems and Control; Graduated with GPA 5.64/6 — *September 2021 - May 2024*
  Courses: *Planning and Decision Making for Autonomous Robots, System Identification, Dynamic Programming and Optimal Control, Robot Dynamics, Image Analysis and Computer Vision, Machine Learning, Deep Learning for Autonomous Driving, Recursive Estimation, Model Predictive Control, Vision Algorithms for Mobile Robotics.*
- **Alma Mater Studiorum - University of Bologna** — Bologna, Italy
  Bachelor in Automation Engineering; Graduated with a final score of 110/110 — *September 2018 - July 2021*
  Courses: *Electronics, Mechanics, Programming (C), Control Systems, Industrial Robotics, Electrical Drives, Computer Architecture.*

Publications

- **Lost in Translation? Vocabulary Alignment for Source-Free Adaptation in Open-Vocabulary Semantic Segmentation (S. Mazzucco, C. Persson, M. Segu, P. L. Dovesi, F. Tombari, L. Van Gool, M. Poggi)**: Paper published at BMVC 2025 (British Machine Vision Conference), based on my Master's Thesis work.

Skills Summary

- **Programming Languages and Frameworks**: Python, Matlab, C++, Pytorch, Git, Linux, ROS
- **Soft Skills**: Team work oriented, time management, critical thinking, effective communicator, reliability and motivation
- **Languages**: Italian (mother tongue), English (C1), German (A2)

Hobbies

- **Sports**: Football: currently playing for FC Italia Zurigo, running (Zürich Marathon 2024, Lisboa Super Half Marathon 2025, NYC Marathon 2025), F1 (worked as a waiter at the 2022 Imola Grand Prix), basketball, tennis.
- **Analogic Photography**: I take pictures with my dad's old analogic Nikon F90X.
- **Travelling**: I had the chance to visit many wonderful countries: Brasil, Thailand, Australia, USA, Indonesia, Costa Rica, Barbados, China, Mexico, South Africa.